

Executive Summary:

Do temporal network signals predict how scientific concepts spread across disciplines?

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September 2026

Research question

Do early structural signals in a concept’s topic co-occurrence network anticipate later cross-field breadth, beyond simple measures of volume and reach?

Data

The study used the OpenAlex bulk snapshot (476M works, 2026-09-23), identified 12,499 concepts by title matching, and tracked 27,393 concept-by-field adoption episodes across 26 venue fields. 53 early network indicators were screened against a five-feature popularity baseline (B5). The analysis used a development set (DEV: 4,771 concepts from CS/Eng/BGM/Med, onset 2003–2009), four held-out field groups (3,372 concepts), a 2010–2014 onset cohort (4,356 concepts), and a 2015–2017 confirmatory cohort (1,443 concepts). A vocabulary-free Frame N (636 phrase-born concepts) provided a further confirmation body.

Approach

Five iterations tested rival mechanisms in sequence. Each iteration commissioned 4–5 artifacts (experiments, evaluations, datasets, or prior-art checks), underwent blocking peer review, and produced a hypothesis update that steered the next iteration. Twenty artifacts were commissioned; sixteen completed and four failed. All five reviews were blocking (scores 2–3).

Key findings

1. **Seven early indicators predict breadth on held-out fields.** Of 53 screened indicators, 10 were frozen on DEV, and 7 were confirmed on held-out groups (Holm $p < 0.05$, CI excluding zero). The confirmed set spans retained-frontier indicators (M0_density_end: +0.375; D_vol_end: +0.307; CONTACT_REACH: +0.211; RETENTION_RATIO: −0.114) and cooccurrence-topology indicators (n_{comm} : +0.167; NOV_{res}: +0.151; ego_density: −0.102). M0_density_end and D_vol_end are partly pre-onset footprint (about 50% attenuation post-onset).
2. **OPEN_{home} is a replicated weak association.** The OPEN index (six signed ego-network components) shows a pooled partial Spearman of +0.069 [+0.038, +0.100] over 6 non-selection

bodies ($I^2 = 0$, 6/6 positive). Fresh-cohort confirmation was marginal: PSP +0.091 at R2, CIs include zero at R4–R5. Frame N: +0.117 at R3 but DL CI includes zero. Predictive value is negligible (+0.002 over B5).

3. **Concepts spread to fields related to those retaining them.** Conditional-logit coefficient $d_0 = 0.322$ [0.291, 0.355] on 6,978 entry events. Verdict: FRONTIER = PARTIAL—the volume-matched contrast was null and d_0 reversed under min-conditional-probability proximity.
4. **The consistency–breadth reversal.** Cheng’s ideational consistency predicts volume (+53.5%/SD) but, net of volume, predicts *narrower* breadth (DL -0.079 , $I^2 = 0$, 5/5 groups negative). Size removes 98% of the volume effect.
5. **Breadth is 73% early exploration, 27% retention.** Log-additive decomposition: early contact diversity contributes 78%, frontier advance -5% , retention 27%.
6. **The signal is topical non-redundancy, not temporal turnover.** Within-concept permutation nulls absorb the year-to-year component. However, turnover could not be tested at available sample sizes.
7. **New-community partners carry the signal.** Partners from new Leiden communities: $C_2 = +0.102$ (Holm $p = 0.0025$); arriving through mixed-field papers: $C_4 = +0.103$ (Holm $p = 0.0025$).

Major dead ends

- Naturalisation gap A_h^* : null ($\Delta\rho = -0.006$), dominated by background homophily ($R^2 = 0.66$).
- Gateway-field retention: held-out $\Delta\text{AUC} -0.00001$; the iteration-1 lead was a domain-specific proxy.
- Within-concept closure: NOT SUPPORTED on DEV, opposite-signed on held-out.
- Discrete trajectory typology: no stable classes (DTW-HMM ARI = 0.222). Spread is a continuum.
- Co-author reach (Candidate S): not confirmed for any outcome.
- Four artifacts failed to execute (Experiment 2, Dataset 1, Experiment 9, Experiment 11 incomplete).

Limitations and open questions

- **Effect size.** The pooled $\text{OPEN}_{\text{home}}$ PSP of +0.069 adds no forecasting value over B5.
- **Mechanism.** The within-concept test was null; the between-concept association could reflect intrinsic generality.
- **Domain heterogeneity.** Life & Environment shows the weakest signal; $I^2 = 0.92$ for the retained-frontier effect across groups.
- **Temporal vs static.** Experiment 16 could not adjudicate temporal churn at ~ 10 papers/year (V2 excess unreliable, planted churn not recovered).
- **No causal claim.** All findings are associational. No instrument or natural experiment was available.

Bottom line

Early cooccurrence openness—specifically, non-redundancy of the home-topic partner mix carried by new-community, mixed-field partners—shows a reproducible but small partial association (+0.069) with later cross-field breadth, beyond a five-feature popularity baseline. The signal replicates across field groups and onset cohorts ($I^2 = 0$), but adds no practical forecasting value and lacks a confirmed within-concept mechanism. Concepts spread to related retained fields ($d_0 = 0.322$), though this is partly confounded with volume. The run’s main methodological contribution is a held-out indicator screen with hash-sealed specifications on a 12,499-concept OpenAlex panel.